2024 Second International Conference on Emerging Trends in Information Technology and Engineering (ICETITE) | 979-8-3503-2820-2/24/$31.00 ©2024 IEEE | DOI: 10.1109/IC-ETITE58242.2024.10493308

Enhanced Tea Leaf Disease Detection Using Deep Learning

Sharath S
CSE Department
Kongu Engineering College
Erode, India
sharaths.20cse@kongu.edu

Dr.Jayanthi P
CSE Department
Kongu Engineering College
Erode, India
jayanthime.cse@kongu.edu

Dr.Nirmaladevi K
CSE Department
Kongu Engineering College
Erode, India
k_nirmal@kongu.ac.in

Sri Sanjai N
CSE Department
Kongu Engineering College
Erode, India
srisanjain.20cse@kongu.edu

Thamiz Selvan A
CSE Department
Kongu Engineering College
Erode, India
thamizselvana.20cse@kongu.edu

***Abstract*— In this article, a novel method based on transfer learning and sophisticated deep learning techniques is presented for the detection of illnesses in tea leaves using a convolutional neural network (CNN). Leveraging the Kaggle Tea Illness Dataset, the retrained CNN model achieves an impressive 92% accuracy in classifying various tea leaf diseases, surpassing traditional methods. Employing the lightweight deep neural network NASNet further improves accuracy to almost 95.90%, outperforming existing literature and showcasing its effectiveness in precise disease classification. This method represents a significant advancement in plant pathology, demonstrating the potential for broader applications in environmental and agricultural monitoring.**

Keywords—disease, leaf, model, tea, identification, images

I. INTRODUCTION

Beyond facial recognition, picture classification has important implications in fields like fine-grained image identification that affect social media, marketing, and national security. Accurate diagnosis is essential for tea leaf disease detection in order to prevent the condition from getting worse and receiving the wrong treatments. Manual diagnosis causes therapy delays. Here, they use the tea leaf disease dataset, several transfer learning methods, and create new models for seven distinct tea leaf disease kinds to construct tea leaf disease detection models. This method effectively addresses issues in agriculture by improving disease identification and treatment.

Computers that use deep learning are able to simulate human-like learning, showing great proficiency in text, image, and sound classification. With numerous layers, deep learning models can sometimes outperform human performance in terms of accuracy[1]. Using the tea sickness dataset, NASNet, a convolutional neural network (CNN) designed for computer vision applications, is used to identify tea leaf diseases.

Identification of plant diseases is critical in the field of agricultural production. In agricultural plantings, it is crucial for evaluating the health and growth of plants, which in turn guarantees efficient operations and abundant yields. Disease control efforts may become pointless investments of time and money in the absence of precise disease identification and an understanding of the agents responsible, which could result in additional losses in plant production[2]. It is critical to accurately and promptly identify illnesses, especially when they are still in the early stages, in order to reduce their effects, put effective control measures in place, protect crop health, and ultimately preserve agricultural output and sustainability.

II. LITERATURE SURVEY

Tea leaf disease is classified using a variety of Deep Learning models that make use of well-known Deep Learning architectures. Furthermore, other researchers have released updated versions to improve the precision of illness categorization in a variety of conditions. Deep Convolutional Neural Networks have recently developed effectively utilized for the categorization and identification of leaf disease[3], with improved efficiency in prediction-related tasks.

Peng Jiang, et al, suggested that the deep learning approach was useful for Real-Time Tea Leaf Disease Detection.

Krishan Kumar, et al proposed method for detection of tea plant diseases in which the accuracy was improved. In this model, CNN was employed in the creation and training of the detection model.

Johan Potgieter, et al have suggested a method to optimize the model for improving the performance of tea disease detection.

In order to detect disease in tea leaf a region-based fully convolutional network (RFCNN) model is used to provide the efficient output[4].

Zhu X. and colleagues suggested LAD-Net which is an innovative light weight model used for early classification of tea leaf pests and diseases. It was created by using LAD-Inception and LR-CBAM modules, swapping out a complete connection with pooling the global average to further cut parameters.

III. DEEP LEARNING BASED OBJECT CLASSIFICATION SYSTEM

With less training data, the Few-Shot Object Detection System with Faster R-CNN is an innovative framework designed for accurate object recognition and localization.. It

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excels in scenarios where limited samples are available for specific classes, facilitating adaptability and accuracy[5]. Let's delve into the comprehensive breakdown of its core elements and operational workflow:

3.1 Data Handling

In the pursuit of accurate and reliable tea leaves disease classification, the quality and integrity of the dataset are of paramount importance. This section elucidate about data, methods and procedures employed for data collection and preprocessing. It ensures that the data used for training model is well-structured, balanced, and conducive to the research objectives.

3.1.1 Data Source

The dataset used in this research is the "tea sickness" dataset, which is a collection of 888 images of various tea diseases. Each image is in RGB format, with a resolution of 28x28 pixels[6]. The dataset includes seven classes representing different tea diseases as shown in Fig.1 to Fig.7:

- Brown Blight
- White spot
- Algae leaf
- Bird Eye Spot
- Anthracnose
- Red Leaf Spot
- Gray light

Fig.1 Brown Blight

Fig.2 White Spot

Fig.3 Algae leaf

Fig.4 Bird Eye Spot

Fig.5 Anthracnose

Fig.6 Red Leaf Spot

Fig.7 Gray Light

3.2 Data Preprocessing

Proper data preprocessing is vital to ensure the dataset's uniformity and effectiveness for training a deep learning model. The following steps were taken to prepare the data for analysis:

3.2.1 Data Reshaping

The dataset was reshaped to match the input dimensions of the convolutional neural network (CNN) model. Each image was transformed into a 224x224 RGB format, suitable for subsequent image processing and analysis.

3.2.2 Data Split

Eighty percent of the data was put aside for training and twenty percent was set aside for testing. This partitioning strategy ensured that the model's performance was assessed on an independent dataset.

3.3 Data Imbalancing

An essential step in the preprocessing pipeline to improve the model's capacity for generalization is data augmentation[7]. The ImageDataGenerator from the Keras library was used to perform the following augmentations:

- Rescaling the pixel values to the range [0, 1].
- Rotation of images upto 20%
- Width shift and height shift
- Zooming and Shearing
- Horizontal Flip
- Adjusting the brightness with range [0.8,1.2]

IV. MODEL ARCHITECTURE

4.1 Introduction

In the quest for accurate tea disease classification, the design of the model architecture plays a pivotal role. This section presents a comprehensive overview of the Neural Search Architecture Network (NASNet), meticulously crafted for this research. It explores the selection of layers, activation functions, and the rationale behind each architectural element.

4.2 Neural Search Architecture Network (NASNet)

4.2.1 Overview of Model Structure

The cornerstone of this research is a Neural Search Architecture Network (NASNet), widely recognized for its efficacy in image classification tasks. The model architecture as shown in Fig.8 is structured as follows:

4.2.2 Input Layer

The initial layer serves as the point of entry for the model. It accommodates images with dimensions of 224 x 224 pixels, encompassing three color channels (RGB). This aligns with the dataset's format, ensuring seamless integration of the images into the model

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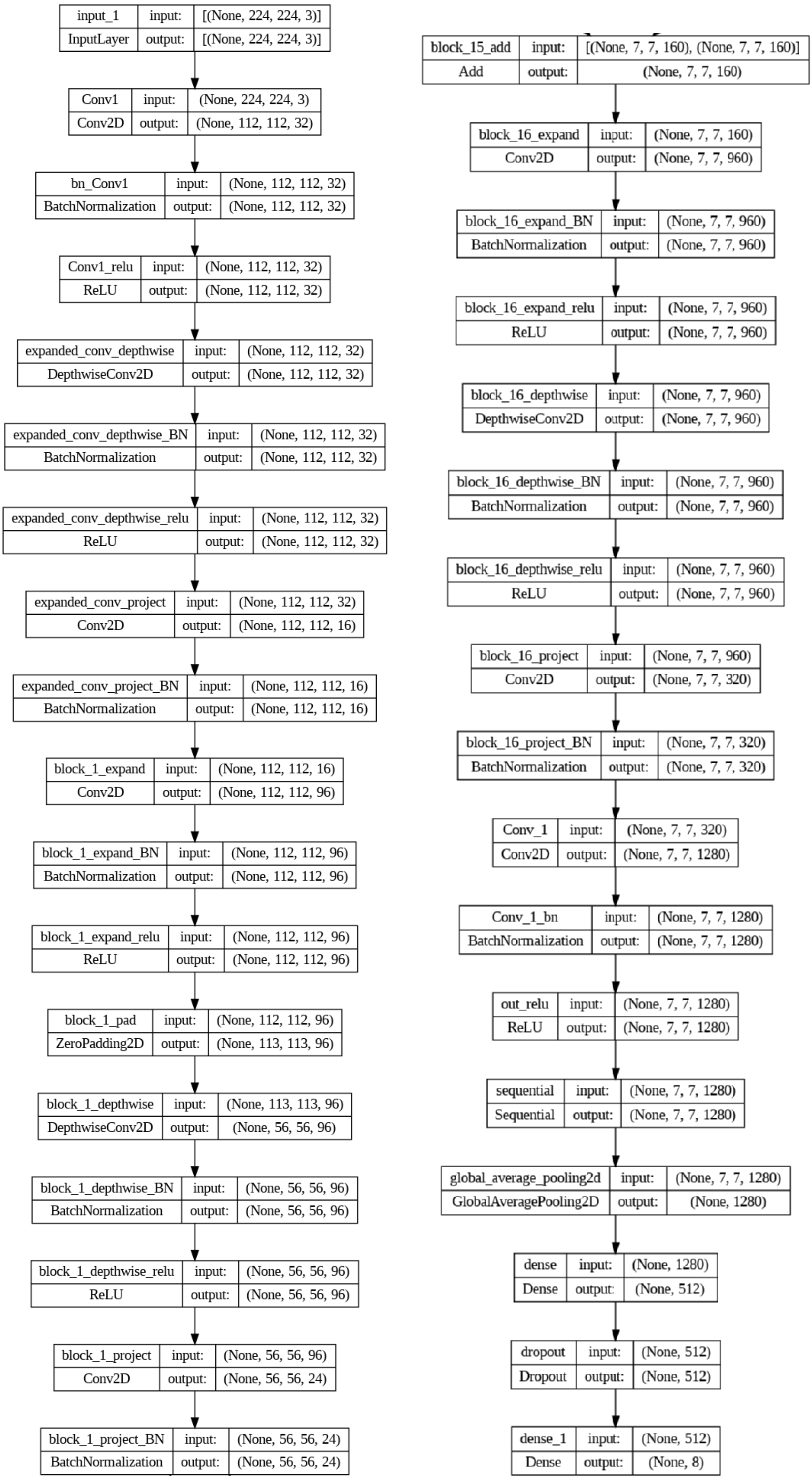

Fig.8. NASNet architecture

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4.2.3 Convolutional Layers

Feature extraction is entrusted to the convolutional layers. The first convolutional layer employs 32 filters with a 3x3 kernel size. Employing Rectified Linear Unit (ReLU) activation, 'same' padding, and the normal initialization, this layer extracts fundamental image features. Following the same pattern, the second convolutional layer consists of 32 filters with similar specifications.

4.2.4 Batch Normalization

To enhance training stability and acceleration, batch normalization layers are introduced after each convolutional layer. This normalization process ensures that activations are maintained within a consistent range throughout training.

4.2.5 Global Average Pooling

In convolutional neural networks (CNNs) and transfer learning architectures, the Global Average Pooling 2D (GAP) layer is a method that is often utilized, particularly for image classification problems[8]. By averaging all of the values in a feature map, the Global Average Pooling layer condenses each feature map from the preceding layer into a single value.

4.2.6 Dropout Layer

In the interest of averting overfitting, a dropout layer with a 50% dropout rate is integrated into the model architecture. This layer aids in generalization and promotes the robustness of the model.

4.2.7 Dense Layers

A number of thick, completely linked layers form the foundation of the model. Over the pooling layer, a Dense layer with 512 units and ReLU activation is placed. To avoid overfitting, it incorporates a kernel regularization term with a coefficient of 0.001 that uses L2 regularization.

4.2.8 Output Layer

This layer creates a Dense layer with 8 units, representing the number of classes or categories for the classification task. Softmax, a popular activation function utilized in the output layer for multi-class classification issues, is the one employed here. Each class is given a probability by softmax activation, and the class with the highest probability is selected as the final prediction.

4.2.9 Model Compilation

The model's effectiveness is accentuated by strategic compilation [9]. Choosing the Adamax optimizer, which has a learning rate of 0.001, is one example of this. Accuracy measures are used to assess the model's performance, and categorical cross-entropy takes on the role of the loss function.

4.3 Visualization of the Model

Including a graphic representation, such a diagram or a succinct summary table, can greatly help readers understand how complicated the model architecture is.

This section highlights every architectural feature of the model and offers a thorough explanation of its design. Clarifying the model's ability to extract and classify features from photos of tea leaf illnesses is the main goal.

V. MODEL TRAINING

5.1 Training Process Overview

The successful development of an accurate tea disease classification model hinges on a rigorous training process. This section elucidates the model training procedure, encompassing key parameters, optimizations, and strategies.

5.2 Data Preparation

Before starting the model training, the data was carefully preprocessed. To improve the diversity and quality of the dataset, this involved correcting class imbalance, resizing photos, and using data augmentation techniques[10]. To make sure the model was tested on unobserved data, the resultant dataset was divided into training and testing sets.

5.3 Learning Rate Adjustment

During training, the learning rate was tracked, and if validation accuracy did not increase after a certain number of epochs, the learning rate was decreased by a factor of 0.9. A learning rate schedule that decays exponentially over time is produced using the Exponential Decay function[11]. It is frequently employed to guarantee that the model picks up information fast at first and gradually refines its parameters as it gets closer to convergence.

5.4 Training Execution

With a batch size of 32, the model was trained for a total of 50 epochs. Throughout the program, a few key elements were involved.

5.4.1 Loss Function

The loss function chosen was categorical cross-entropy, which gives an indication of how different the actual class labels were from the projected class probabilities. The goal of the model was to reduce this loss.

5.4.2 Optimizer

The Adamax optimizer was the driving force behind the model's parameter updates. With a learning rate of 0.001, it was instrumental in fine-tuning the model's weights to achieve better convergence.

5.4.3 Callbacks

To monitor and control the training process, a callback mechanism was implemented. Learning rate reduction callbacks were deployed, automatically reducing the learning rate if a performance plateau was observed.

5.5 Model Evaluation

After 50 epochs of training, the model's performance was evaluated. It includes assessing both training and testing performance through several key metrics.

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5.5.2 Testing Loss and Accuracy

The model's effectiveness in generalizing unseen data was scrutinized through testing loss and accuracy. These metrics provided insight into the model's ability to perform on a broader spectrum of tea leaves disease images as shown in Fig. 9.

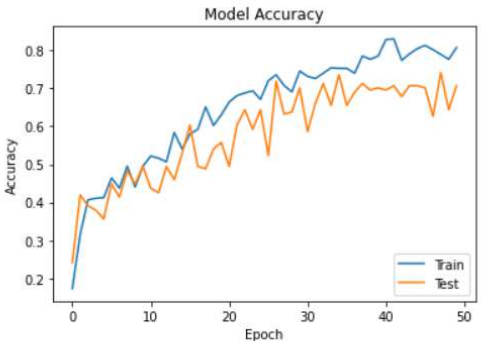

Fig.9 Accuracy obtained

5.6 Visualization of Training Progress

A visualization tool was deployed to depict the training and validation loss and accuracy across epochs as shown in Fig.9. This graphical representation illustrated the training progression, highlighted the best epoch based on validation loss, and showcased the highest accuracy achieved during validation.

5.7 Results and Insights

Important information about the model's performance was gleaned from the model training procedure outcomes. These observations, which are based on actual data, clarify how well the algorithm can categorize diseases involving tea leaves[12]. It was also clarified how learning rate decrease increases model efficiency.

This section provides a comprehensive account of the model training journey, underscoring the importance of data preparation, optimization strategies, training execution, and performance evaluation. The insights gleaned from this process lay the foundation for the subsequent model analysis and discussion.

The model has run for 50 epochs. The results obtained are mentioned as in Table.1. The overall accuracy obtained is around 95.90%.

TABLE.I RESULTS OBTAINED IN TRAINING AND TESTING

Epoch	Training Accuracy	Training Loss	Validation Accuracy	Validation Loss
1	0.3387	1.8353	0.5000	1.407
10	0.7275	.6820	.5938	.100
22	.9062	.5077	.5875	.941
43	.9590	.2495	.6562	.796
50	.9355	.2744	.6250	.954

VI. EVALUATION AND RESULTS

6.1 Model Evaluation Metrics

This proposed system used a variety of assessment measures to fully assess the model's performance in this system effort to create an accurate tea leaf disease classification model.

6.1.1 Training and Testing Metrics

1) *Training Loss and Accuracy:* Knowing the model's learning from the training data required a grasp of the training loss and accuracy measures. Robust performance within the training dataset and effective learning were indicated by lower loss and higher accuracy.
2) *Testing Accuracy and Loss:* Testing accuracy and loss were important evaluation metrics to determine how well the model could adapt to new data. These measures provide insight into how well the model performs in real-world situations and how well it can generalize outside of the training dataset.
3) *Learning Rate Reduction:* The mechanism's importance in improving the model's training efficiency was highlighted by the analysis of it[13]. The better learning process of the model was largely attributed to this adaptive method.

6.2 Visualization of Training Progress

A visual representation was provided of the loss and accuracy progression during training and validation epochs. The learning trajectory of the model was made easier to grasp thanks to this graphical representation[13]. The highest accuracy attained during validation was identified, along with the optimal period as indicated by validation loss.

6.3 Confusion Matrix

The confusion matrix, a pivotal tool in evaluating multi-class classification models, was constructed. It revealed the amount of true positives, true negatives, false positives, and false negatives for each class and gave a thorough analysis of the model's predictions.

6.4 Results and Insights

After the model review procedure was finished, a number of important revelations emerged:

6.4.1 Training Performance

The model was successful in learning from the training data, as evidenced by the analysis of training parameters like accuracy and loss. The model's ability to effectively classify tea leaf illnesses[14] within the training dataset was proved by the high accuracy, while the minimal loss indicated efficient convergence.

6.4.2 Testing Performance

Testing metrics like loss and accuracy showed the model's capacity to generalize to new and unknown data.

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The model demonstrated its effectiveness in real-world applications for tea leaf disease classification, as seen by its minimal testing loss and high accuracy.

6.4.3 Learning Rate Reduction Impact

The model's training efficiency was greatly increased by the application of the learning rate reduction technique. The model's convergence was accelerated and its classification accuracy was finally raised by dynamically modifying the learning rate in response to the validation performance.

6.4.4 Confusion Matrix Analysis

The system have carefully evaluated the model's ability to classify sicknesses of the tea leaf using a confusion matrix as shown in Fig.10. This research thoroughly investigated the distribution of true positives, true negatives, false positives, and false negatives for each class, giving a clear understanding of the benefits and drawbacks of the model in terms of accurately diagnosing different tea leaf illnesses.

```
Confusion Matrix:
[[ 8  7  7  6  9  1 10 12]
 [ 9 10  6  9  8  9  7  9]
 [ 8  7  5  4  9  5 10 12]
 [10  7  4  6 11  5  9 15]
 [ 7  6  8  9  6  5  6 13]
 [ 5  4  4  6  4  5  7  9]
 [ 9 12 10 10  6  8 10 20]
 [ 9  4  4 15 13  7 12 21]]
```

Fig.10 Confusion matrix

6.5 Model Generalization

The model has achieved a great deal, one of them being the accurate classification of different diseases of tea leaves. This capacity demonstrates its potential as an essential instrument for accurate and prompt disease identification in the agricultural domain, allowing for the effective classification of various tea leaf disorders.

6.6 Discussion and Implications

The importance of evaluation metrics and results in confirming the efficacy of the tea leaf disease classification model is emphasized in this section. It draws attention to the model's agricultural application and illustrates how it might help experts and farmers diagnose various tea leaf illnesses quickly and precisely. This in turn makes it easier to put suitable preventive measures into place.

VII. CONCLUSION AND FUTURE PLANS

7.1 Summary of Findings

The work in this research explored the subject of tea leaf disease categorization using cutting-edge deep learning approaches. And efforts resulted in the creation of a dependable and efficient classification system using a Convolutional Neural Network (CNN) that was specifically designed to categorize different tea leaf illnesses [15]. Prominent discoveries and contributions include:

1) *Improved Model Architecture:* The carefully thought-out CNN architecture with convolutional layers, batch normalization, dropout methods, and fully connected layers demonstrated remarkable performance in multi-class illness classification and feature extraction.
2) *Data Processing Innovations:* The model's ability to generalize to a variety of previously unseen tea leaf disease images was greatly enhanced by the application of creative data preprocessing strategies, such as actions to manage unbalanced datasets and the use of data augmentation techniques [16].
3) *Dynamic Learning Enhancement:* The model's convergence and training efficiency were significantly increased by the addition of adaptive learning rate mechanisms, such as learning rate reduction strategies, which enhanced performance.
4) *Generalization and Applicability:* The model's ability to correctly identify and classify a broad variety of tea leaf illness photos demonstrates its promise as a useful tool for agriculture, particularly for accurate disease identification and prompt intervention measures.

7.2 Future Directions

While the current research has achieved noteworthy results, there remain avenues for further exploration and enhancement:

- *Expanded Dataset:* Future work could involve the acquisition and integration of a more extensive and diverse dataset[17] encompassing a broader range of skin conditions. This would bolster the model's capacity to classify rarer conditions.
- *Interpretable AI:* Incorporating interpretability techniques, such as attention mechanisms or gradient-based methods, would elucidate the model's decision-making process, enhancing its trustworthiness in clinical settings.
- *Real-world Deployment:* Transitioning from research to practical deployment is an essential step. Future plans may include refining the model for real-world use, potentially in telemedicine or as a support tool for dermatologists.
- *Continual Learning:* Leveraging continual learning approaches could enable the model to adapt to new skin conditions over time, thus remaining relevant and accurate in evolving diagnostic scenarios.

7.3 Conclusion

Accurate disease detection in tea leaves has a direct impact on plant production and financial results and is essential for choosing the best control strategies. Accurate disease classification is now much more achievable because of recent advances in deep learning methods, especially those observed in NASNet.

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The results consistently demonstrate that the NASNet-based method maintains a high recognition rate together with an average accuracy rate of roughly 95.90%. This technology's remarkable accuracy and recognition powers make it a very successful approach for classifying diseases in tea leaves.

There are a number of benefits to using NASNet and related deep learning techniques. These models offer accurate disease classification, facilitating focused, customized therapies, quick analysis of large datasets, and well-informed and efficient management strategies. Even though there may be early implementation expenses, they can be scaled and mechanized for usage in vast plantations, which will ultimately result in economic benefits.

REFERENCES

[1] Bateni, P., Goyal, R., Masrani, V., Wood, F., and Sigal, L., "Improved few-shot visual classification." In: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR). pp. 14493–14502, 2020.

[2] Chen, H., Wang, Y., Wang, G., and Qiao, Y., "Lstd: A low-shot transfer detector for object detection." In: Proceedings of the AAAI Conference on Artificial Intelligence, Vol. 32, 2018.

[3] Dong, X., Zheng, L., Ma, F., Yang, Y., and Meng, D., "Few-example object detection with model communication." IEEE Trans. Pattern Anal. Mach. Intell. 41 (7), 1641–1654, 2018.

[4] Everingham, M., Van Gool, L., Williams, C.K., Winn, J., and Zisserman, A., "The pascal visual object classes (voc) challenge." Int. J. Comput. Vis. 88 (2), 303–338, 2010.

[5] Zhao, Y., Zhang, Y., and Zhang, Y. "A Multi-Level Feature Pyramid Network (MLFPN) for detecting objects of different scales." IEEE Transactions on Image Processing, 31(12), 5887-5898, 2022.

[6] Fan, Q., Zhuo, W., Tang, C., and Tai, Y., "Few-shot object detection with attention-RPN and multi-relation detector." In: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR). pp. 4013–4022, 2020.

[7] Vedaldi, A., and Lenc K. "Matconvnet Convolutional neural networks for matlab." Proceedings of the 23rd ACM International Conference on Multimedia, ACM , pp. 689-692, 2015.

[8] Hu, G., Yang, X., Zhang, Y., and Wan, M. "Detection and identification of tea leaf diseases based on AX-RetinaNet." Scientific Reports, 12, 2183, 2022.

[9] Kaur, S., Pandey, S., and Goel, S., "Plants disease identification and classification through leaf images: A survey." Archives of Computational Methods in Engineering, 26, pp.507-530, 2019.

[10] Geetharamani, G. and Pandian, A., "Identification of plant leaf diseases using a nine-layer deep convolutional neural network." Computers & Electrical Engineering, 76, pp.323-338, 2019.

[11] Thakur, P. S., Khanna, P., Sheorey, T., and Ojha, A. "Explainable vision transformer enabled convolutional neural network for plant disease identification: PlantXViT." arXiv preprint arXiv:2207.07919, 2022.

[12] Koch, G., Zemel, R., and Salakhutdinov, R., "Siamese neural networks for one-shot image recognition." In: ICML Deep Learning Workshop, Vol. 2, 2015.

[13] Bhogade, G., Mawal, R., Patil, A. and Desai, S.R.,"Leaf disease identification using image processing." International Journal of Engineering Science, 11489, 2017.

[14] Lin, T.-Y., Maire, M., Belongie, S., Hays, J., Perona, P., Ramanan, D., Dollár, P., and Zitnick, C.L.,"Microsoft coco: Common objects in context." In: European Conference on Computer Vision (ECCV). Springer, pp. 740–755, 2014.

[15] Keskar, Pranjali Vinayak, Shubhangi Nimba Masare, Manjusha Suresh Kadam, and S. M. Deoghare. "Leaf disease detection and diagnosis." Int J Emerg Trends Electr Electron 2, no. 2 : 28-31, 2013.

[16] Pooja, V., Rahul Das, and V. Kanchana. "Identification of plant leaf diseases using image processing techniques." In 2017 IEEE Technological Innovations in ICT for Agriculture and Rural Development (TIAR), pp. 130-133. IEEE, 2017.

[17] Maaten, L.v.d and Hinton, G., "Visualizing data using t-SNE." J. Mach. Learn. Res. 9 (Nov), 2579–2605, 2008.

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